



AFLA BEGINNER GUIDE 3 OF 3

Somatic (tumour) analysis

Low-frequency variants, UMIs and amplicons, tumour-only vs paired, TMB, MSI, evidence and tiers

Education and research use only · not for clinical diagnosis or patient management

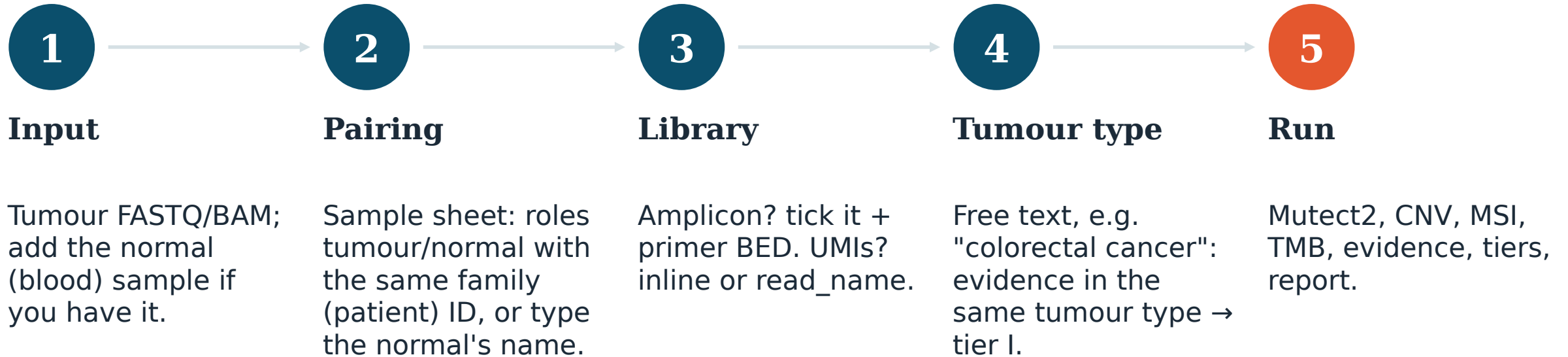
Germline vs somatic

Same reads, different questions

	Germline	Somatic
Where the variant is	every cell, from conception	only in tumour cells
Expected VAF	≈ 50% (het) or 100% (hom)	anything: 1-60%, depends on purity and clonality
Caller	GATK HaplotypeCaller / DeepVariant	GATK Mutect2
Depth needed	≥ 20× per base	often ≥ 200-500× for panels
Main risk	missing a variant in low coverage	calling artefacts or germline variants as somatic
Interpretation	ACMG/AMP pathogenicity	AMP/ASCO/CAP tiers (clinical actionability)

Running a tumour

EPI2ME → AFLA somatic



Amplicons and UMIs, simply

Two chemistries that change the analysis



Primer clipping

Primer bases copy the oligo, not the patient. AFLA soft-clips them using the primer BED, so they cannot hide or fake variants at amplicon ends. In a test, 335,038 reads were clipped with 98% amplicon uniformity.



UMI consensus

Each original DNA molecule carries a barcode (UMI). Reads with the same UMI are merged: random PCR and sequencing errors disappear. In a simulation, AFLA recovered the planted families of 1-4 reads exactly.

- Read structures: "8M+T 8M+T" = 8-base UMI on both reads; "+T 12M11S+T" = QIAseq (UMI + spacer on read 2).
- Illumina manifest → primer BED: `python3 scripts/manifest_to_primers.py manifest.txt mypanel`

Tumour-only or tumour + normal?

The normal sample is the best germline filter



Tumour + matched normal

Variants also in the normal are inherited, not tumour. Most reliable; needed for accurate TMB.



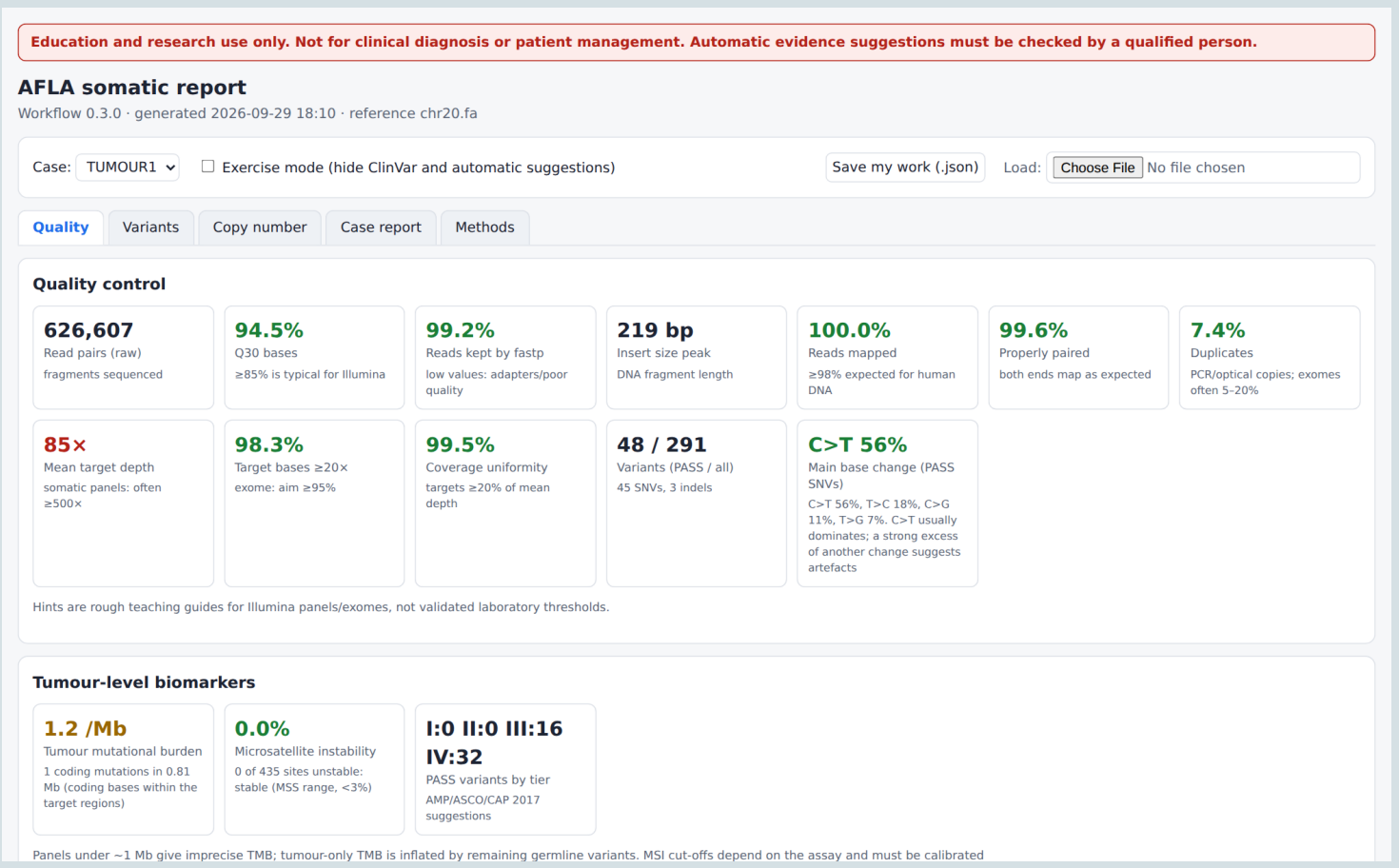
Tumour-only

Germline variants must be guessed from gnomAD, the panel of normals and VAF. AFLA shows an "origin hint": somatic candidate / possible germline (VAF) / likely germline (population).

Practice data: an in-silico "tumour" made by mixing 20% of one person's reads into another's. Their private variants appear at $\approx 10\text{-}15\%$ VAF, and common polymorphisms are correctly rejected as germline.

Tumour-level biomarkers

TMB and MSI in the Quality tab



Tiers: how actionable is a variant?

AMP/ASCO/CAP 2017, suggested by AFLA, decided by you



Strong clinical significance

Level A/B evidence (CIViC) or OncoKB level 1/2 in THIS tumour type



Potential clinical significance

A/B in another tumour type, level C/D, OncoKB 3/4, recurrent COSMIC hotspot



Unknown significance

No evidence found, rare in the population



Benign or likely benign

Common in the population ($\geq 1\%$) or ClinVar benign

Variants tab (somatic)

Tier, origin hint and evidence in one view

Education and research use only. Not for clinical diagnosis or patient management. Automatic evidence suggestions must be checked by a qualified person.

AFLA somatic report

Workflow 0.3.0 · generated 2026-09-29 18:10 · reference chr20.fa

Case:

TUMOUR1

☐ Exercise mode (hide ClinVar and automatic suggestions)

Save my work (.json)

Load:

Choose File

No file chosen

Quality

Variants

Copy number

Case report

Methods

Variant explorer

Tier I-II

Somatic candidates (coding)

All variants

Quality

☒ PASS only

Min depth

0

Min VAF

0.02

Population frequency

Max gnomAD / max pop. AF

0.001

☒ keep variants with no known frequency

Predicted impact (VEP)

☒ HIGH
☒ MODERATE
☐ LOW
☐ MODIFIER
☒ + predicted splice effect

Somatic

☒ somatic candidates only

Tier

any

ClinVar

☐ pathogenic / likely pathogenic only

Genes

or type genes: BRCA1, BRCA2, PALB2

Search

gene, HGVS, rsID...

3 of 291 variants shown

Click a column to sort, a row for details. ☆ adds a variant to the case report.

Export filtered rows (CSV)

☆	Gene	Variant	HGVSp	Consequence	VAF	DP	Origin (hint)	Tier	Evidence	gnomAD / max AF	ClinVar	Filter
☆	SMOX	chr20:4182414 C>T	p.Ser312Leu	missense variant	0.03	181	somatic candidate	III		4.9e-5		PASS
☆	POFUT1	chr20:32234536 G>A	p.Asp348Asn	missense variant	0.18	82	somatic candidate	III		7.6e-4		PASS
☆	ZNF512B	chr20:63964672 G>A	p.Ala360Val	missense variant	0.04	154	somatic candidate	III		-		PASS

< Prev

page 1 / 1

Next >

Origin hint somatic candidate vs likely germline.

Tier hover to see why.

Evidence CIViC items, OncoKB level, COSMIC count.

Details click a row: evidence table with therapies and links.

Presets: Tier I-II, Somatic candidates, All

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Evidence sources

All free for education; some need registration



CIViC

Open (CC0) expert-curated cancer evidence: therapies, diagnosis, prognosis. Downloaded by `afla-setup.sh`.



OncoKB

Precision oncology knowledge base. Free academic token; queried online during the run.



COSMIC

How often a mutation is seen in cancers. Free for academic use after registration; you download it.



ClinVar somatic

Oncogenicity and clinical-impact classifications (ONC/SCI fields) in the same ClinVar file.

Copy number and fusions in tumours

Amplifications, deletions, rearrangements



CNV (CNVkit)

Gains/losses vs the normal (or other normals). Strong amplifications/deletions are matched to CIViC (e.g. ERBB2 amplification). Purity dilutes log2 ratios.



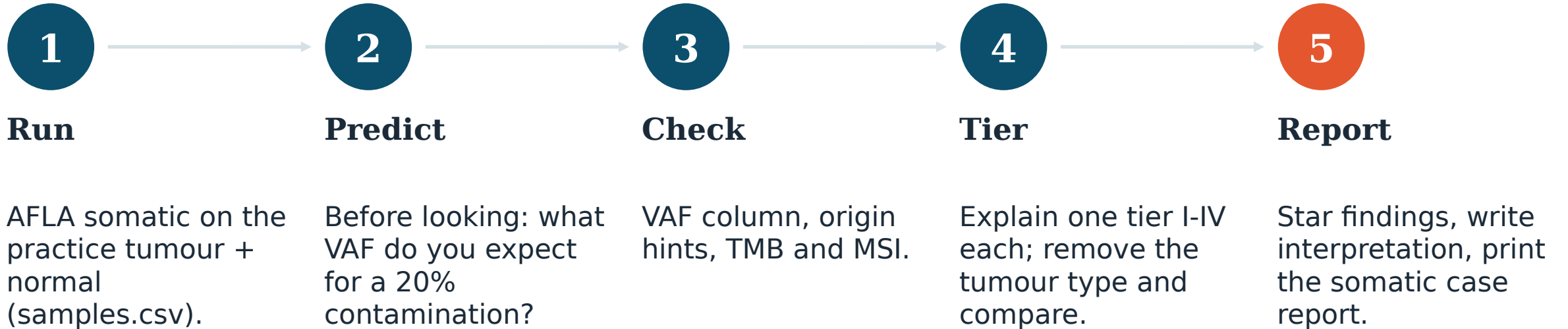
Fusions (Manta)

Translocations between two genes appear as fusion candidates, only if the breakpoint lies near baited regions. Confirm by RNA or FISH.

- Exome/panel CNV is screening-level: confirm clinically relevant changes by another method.
- RNA fusion panels (e.g. anchored multiplex PCR) are not supported yet.

Practice exercise

Work it through, then discuss





Somatic pitfalls to remember

- Tumour-only cannot prove a variant is somatic: consider a germline test for hereditary cancer genes.
- Low VAF + one dominant base change or strand-bias = suspect artefact (FFPE, PCR).
- Purity matters: a 30% tumour gives $\approx 15\%$ VAF for heterozygous clonal variants.
- TMB and MSI thresholds are assay-specific; panels under 1 Mb are imprecise.
- Tiers are suggestions from databases; actionability needs a tumour board.